

STATEMENT OF WORK (SOW)

FOR

Group: 309 SOFTWARE ENGINEERING GROUP (SWEG)

Program: 521AB Polyalphaolefin (PAO) Chiller System

Project/Requirement Title: F-35 Program

Location: Building 1513, Hill Air Force Base, UT

Project Title: Install PAO Chiller System

Hill Air Force Base (HAFB), Utah



January 21, 2026

Version 1

SOW Modification History

Date	Change	Paragraph #

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1.0 INTRODUCTION

1.1 Purpose

Provide and install new PAO chiller, pump, manifolds, distribution piping, controls, and instrumentation in building 1513.

1.2 Scope

The Contractor shall provide a complete product and service that is ready for immediate use in all areas of the project. The Contractor shall deliver all items to the final resting place of the PAO Chiller system. When the installation is complete and final inspection has been accomplished by SWEG Facility Engineering and the assigned Contractor Officer Representative (COR), all deliverables shall meet all codes and government specifications found in this SOW.

The Contractor shall comply with Uniform Facilities Criteria (UFC) (available at wbdg.org), International Building Code (IBC) Codes, National Fire Protection Association (NFPA) Codes, National Electric Code (NEC) Codes, International Mechanical/Plumbing Codes, Hill AFB Installation Facility Standards (IFS), American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) Standards, American Water Works Association (AWWA), the 75th Comm Squadron TAB K (latest edition) and all manufacturers recommend installation, start-up and operation instructions. For any document referenced, the version of the code to adhere to shall be the version of the code in effect at time of proposal. A representative on the SWEG Facility Engineering team will verify the latest version of the code is being utilized.

1.3 Background

The Contractor shall perform the service required after coordinating with Facility Project Manager (PM). Normal working hours are 0600 to 1700 Monday through Friday. Saturday work is only available with prior approval. No overtime costs shall be allowed. Any work that needs to be done outside of working hours must be approved. The General Special Access Program (SAP) Security Officer (GSSO) with the Advanced Program Office (APO) must approve such work outside of normal working hours and must ensure escort availability in addition to the building being accessible at all times.

2.0 DESCRIPTION OF WORK/OBJECTIVES/SERVICES

<u>DESCRIPTION</u>	<u>QTY</u>
Chilled water (Fixed) PAO skid system	<u>2</u>
Manifold with remote HMI panel	<u>6</u>
Crating, vacuum packing and transportation of PAO Chiller skids.	<u>2</u>
Crating, vacuum packing and transportation of Manifolds.	<u>6</u>
Skid commissioning and staff training for two Engineers at site	<u>2</u>
Site installation of redundant Chilled water PAO system,	<u>1</u>
Quick Disconnects (QDs)	<u>48</u>

- 2.1 Hard or rigid stainless-steel piping must be used to distribute PAO fluid from the PAO pumps to the distribution manifolds. Any changes must be approved by the SWEG Facilities Engineering team. The Contractor shall use Swagelok fittings and valves to change direction or terminate or start line sets. Mandrel bent radiuses must be used where feasible to maintain fluid flow characteristics and reduce the number of fittings and seams needed.
- 2.2 Contractor shall provide Safety Data Sheets (SDS) for all Hazardous Materials (Hazmat) used.
- 2.3 Contractor shall provide to the COR hard and soft copies of design files which shall be considered government property upon delivery and can be used for government purpose rights (CDRL A001).
- 2.4 Submittals Required: Complete to scale, ¼" equals one (1) foot. Engineered and stamped designs for PAO Chillers, pumps, all associated piping for PAO distribution, and piping for chilled water. Approval required from SWEG Facilities Engineering team before construction. (CDRL A001, A002)
- 2.5 Should the Contractor need to relocate any existing equipment for this project, they shall first notify the SWEG Facilities Engineering team and obtain prior approval. The Contractor is responsible for ensuring all moved equipment functions properly and for repairing any damage that occurs during relocation. Unless identified beforehand, all equipment will be considered in good working order prior to being moved. After the Contractor returns the

equipment to its original place, the government will be responsible for verifying it is fully operational.

2.6 Work area must be properly cleaned and maintained. It is the contractor's responsibility to remove waste generated by this project. Facility dumpster usage must be allowed for day-to-day refuse. Larger waste must not be allowed in HAFB dumpsters and must be removed from site by the Contractor at the end of each workday.

2.7 This document describes six (6) manifolds in rooms B102 (2), B200 (2), A201(1) and A204(1). The manifolds described are new equipment, designed to be low-profile wall-mounted PAO control centers that provide independent flow, pressure regulation, and specifically tailor the pressure and/or flow parameters for each cooling node. They display chilled water inlet and outlet pressure and temperature, PAO inlet and outlet pressure and temperature, and they contain the pump status screens.

2.8 Every manifold contains requisite pressure regulators, aircraft quick-connects, valves and maintenance shut off valves. Every unit is equipped with emergency shutoffs that sever power to the operational pump. Except for emergency stops, only the manifold in room B102 has the capability to power the pump skids on and off. Pressure regulation is required at each manifold such that it can be adjusted depending on the equipment connected to the system. This must prevent over pressurizing causing damage.

2.9 New PAO supply and return lines shall connect the newly designed manifolds to the PAO distribution blocks in rooms: B102, B200, A201 and A204. Include Parker-Eaton Aeroquip 3200 Series Aircraft Fittings/QDs where feasible on the PAO distribution blocks and manifolds. The Parker-Eaton Aircraft Fitting style is noted in Appendix C. Any changes to the fittings must be approved by 309 SWEG 521 SWES. Below are requirements the government has for each half of the connectors:

2.9.1 Eaton Quick Disconnect Coupler, -16 size, 1-5/16" -12 thread, without relief valve, Aluminum

2.9.2 Eaton Quick Disconnect Nipple, -16 size, 1-5/16" -12 thread, without relief valve, Aluminum

2.10 Contractor shall be a U.S. citizen, provide driver's license number, driver's license issuing state, name, and birthday for any personnel that must be performing work on site, at least two (2) weeks in advance of arrival date given to the SWEG Security office. Any exception to this policy shall be upon the government's discretion and not be considered as an establishment of a precedent. The general Contractor shall be responsible to work with 75th SFS and provide all required documentation to obtain

access to Hill AFB. Once base access is granted, identification and proof of citizenship must be required to gain access to the facility where the work must be performed.

2.11 Contractor shall schedule working days at least two (2) business days in advance. Any exception to this policy shall be upon the government's discretion and not be considered as an establishment of a precedent.

2.12 If the Contractor does not show up to the agreed upon location within 15 minutes of appointment they must not be allowed to work that day. Any exception to this policy must be upon the government's discretion and not be considered as an establishment of a precedent.

2.13 Contractor shall stay with an escorting official at all times when in classified areas. Any personnel working on site shall wear high visibility vests or shirts.

3.0 PROJECT TASK/ITEM DESCRIPTION

3.1 The pump and chiller equipment shall be no larger than 121 sq. ft. and must be able to fit through double doors measuring 84 in. tall and 70 in. wide.

3.2 Existing facility Glycol chilled water shall be used as the cooling medium. A supplemental refrigeration system may be used if required.

3.3 The system shall be a redundant chiller design, comprising of pump skid A and pump skid B. Each skid shall be capable of managing a 200 kW heat load. Pump skids A and B must stay connected to power, chiller water plumbing, and PAO oil plumbing. The new chiller design shall allow for automatic repositioning of the system's water and PAO fluid flow control and positioning of the electrical switches in the utility room from pump skid A to pump skid B and vice versa.

3.4 Each chiller skid shall be a fully independent, self-contained chiller unit capable of delivering chilled PAO fluid at 20 GPM between 120 and 130 PSI at 59 degrees Fahrenheit, ± 3 degrees. Chilled PAO fluid shall be delivered to each of the six (6) manifolds. Chilled water and PAO connections installed by the Contractor to the facility shall be independent valves. Chiller connections shall be accessible and removable for equipment change-outs.

3.5 Each of the six (6) manifolds must be capable of simultaneously supplying PAO at 20 GPM at 130 PSI after parasitic losses have been factored.

3.5.1 Each manifold must be adjustable from 0 PSI to 130 PSI, and 0 GPM to 20 GPM.

- 3.6 Each manifold must behave independently and cannot be affected by the flow, pressure, or temperature of adjacent manifolds connected to the same system.
- 3.7 All PAO and water supply lines must be insulated. All lines connecting the manifolds to the pump must be stainless steel hard lines. All chilled water lines must be hard lines. Fiberglass Insulation, ASTM C552, Type II, and Type III, with All-Service Jacket. 1-1/2" Thickness, installed with all longitudinal overlaps sealed and all circumferential joints ASJ taped.
- 3.8 Each manifold shall have independent, immediately accessible, manually operated shut off valves. The Contractor shall verify the final location with the customer prior to installation.
- 3.9 The system shall start with a single action on the control panel utilizing a button press or switch flip.
- 3.10 The system shall stop with a single action on the control panel utilizing a button press or switch flip.
- 3.11 The system shall have an emergency stop function.
- 3.12 The system shall have a user programmable PAO temperature set-point.
- 3.13 The system shall have a safety and alarm system with the following specific alarms to protect the PME cooling loop on each of the six (6) manifolds and each of the two (2) pump skids:
- 3.13.1 High Pressure Functionality - If the PAO supply pressure is above 130 PSI, then the chiller shall shut down without delay and trigger a visual and audio alarm on the remote-control panels and at the chiller.
 - 3.13.2 Low Pressure Functionality - If the PAO supply pressure is below 25 PSI, then the chiller shall shut down without delay and trigger a visual and audio alarm on the remote-control panels and at the chiller.
 - 3.13.3 High Temperature Functionality - If the PAO temperature is above a user-specified temperature then the chiller shall trigger a visual and audible alarm on the remote-control panels and at the chiller.

- 3.13.4 Fluid Reservoir Level - If the reservoir level drops below a safe level, then the chiller shall shut down without delay and trigger a visual and audible alarm on the manifold remote control panel screens and at the chiller.
- 3.13.5 Spill and overfill alarm indication must be displayed on each manifold control screens.
- 3.13.6 Variance from user specified temperature – user shall be able to adjust a variance specification for the allowable temperature tolerances from the user set point. If the temperature is outside of the adjusted tolerance for more than three (3) minutes, then the chiller shall trigger a visual and audible alarm on the remote-control panels and at the chiller.
- 3.13.7 Any alarm condition shall be able to be reset and silenced from the chiller or remote-control panels. When an alarm is reset, the visual and audible cues shall cease.
- 3.13.8 The user shall have the ability to view the chiller PAO input pressure, PAO output pressure, chilled water input pressure, chilled water output pressure, PAO input temperature, PAO output temperature, chilled water input temperature, chilled water output temperature, PAO output flow, and PAO input flow via the remote control panels and at the chiller.
- 3.14 The safety and alarm system shall distinguish which alarm(s) is/are triggered.
- 3.15 The safety and alarm system shall retain a user accessible, time stamped log of alarms.
- 3.16 The chiller skid equipment installation shall have an internal, drainable spill and leak containment system to contain all of the fluid in an approved container in the event of a spill or leak.
- 3.17 The chiller skid equipment installation shall have an access point to drain the spill containment system into an external container. Spill containment must be designed to contain the potential fluid in the system and oil contained in the skid reservoir in operation. The drain system must contain a valve and be properly marked where spill collection containers are located.
- 3.18 The system shall have the ability to continuously separate water from the PAO fluid utilizing a filter.
- 3.19 Pro-Press fittings shall not be used anywhere on the system

- 3.20 Approximate flow velocity calculations shall be provided before commencement of installation. Industry flow velocity standards shall be followed. The Contractor shall follow the most current and most relevant standard.
- 3.21 IAW Installation Facility Standard (available at wbdg.org) chiller system and pumps must be properly braced to prevent damage during an earthquake. See appendix C.
- 3.22 The PAO chillers and pumps must be installed on a seismically engineered support structure that incorporates spill containment, compliant with all applicable building codes. The government requires certified and stamped drawings from a registered seismic engineer demonstrating that the support system meets seismic requirements and effectively isolates the building structure from unacceptable vibration levels, ensuring both occupant comfort and structural integrity.
- 3.23 Chiller system must have an expansion capability that allows the user to upgrade the system in the future
- 3.24 Aerospace grade fittings (Eaton Aerospace or approved equivalent) must be used at points where hoses connect and disconnect. All other portions shall be run in permanent hard line.
- 3.25 The vendor shall provide new flexible 1" stainless steel braided lines to each sensor manifold nodes.
- 3.26 In addition to the regulator, ball valves shall be installed at every manifold supply and return Aline to physically control PAO distribution.
- 3.27 Chiller and pump must be installed on risers to place them at the same level as the raised floor to accommodate ease of access.
- 3.28 Pumps shall only be turned on in one (1) location in room B100, not at every manifold. This precaution is required to avoid unintended test disruptions.
- 3.29 System shall have shoring and a drainable oil splash basin.
- 3.30 There shall be a pressure regulator at each manifold (PSIg)
- 3.31 Contract Data Requirements List (CDRL)

3.32 Contractor shall be compliant with the following guidance: Intelligence Community Directive (ICD 705), Joint Special Access Program Implementation Guide (JSIG), and National Security Council Instruction (NSCI) 7700. For any document referenced, the version of the code to adhere to shall be the version of the code in effect at time of proposal and will be verified by the GSSO in the APO. Some Advanced Program Office requisites require:

- 3.32.1 No Special Access Program Facility (SAPF) penetration must exceed 96 square inches
- 3.32.2 All penetrations in SAPF spaces must be fire caulked after service mediums have been installed
- 3.32.3 Blue OR Green Shielded Category 6 plenum grade copper data cable must be used for controller-pump communication
- 3.32.4 Emergency "E-stop" buttons that terminate power to the pump must be copper transmission lines
- 3.32.5 All services must have easily identifiable labels immediately inside and outside the SAPF penetration locations
- 3.32.6 All cables shall be run in metal conduit
- 3.32.7 The section of conduit penetrating the SAPF wall must be installed and surface mounted with electrically nonconductive material.

3.33 Contractor shall provide a full technical documentation package to include, but not limited to:

- 3.33.1 Building Modifications
- 3.33.2 Equipment Lists
- 3.33.3 Fabrication Drawings
- 3.33.4 General Arrangement Drawings
- 3.33.5 Grounding Details
- 3.33.6 Instrument Locations
- 3.33.7 Layouts
- 3.33.8 Maintenance Manual
- 3.33.9 Panel Drawings
- 3.33.10 Piping Details
- 3.33.11 Red line drawings (sent at the end of the project)
- 3.33.12 User's Manual
- 3.33.13 Wiring Diagrams

Drawings/details shall be submitted to the COR and SWEG Project Engineer (PE) within 30 days after award of contract. Drawings shall be in AutoCAD .dwg and/or .pdf format. Product literature shall be submitted in .pdf format. No work shall proceed without

approved submittals. Government PE will have 14 days to review drawing and accept or reject. If initial submittals are rejected, Contractor shall have 14 calendar days to re-submit and Government PE shall have 7 days to review.

- 3.34 Contractor shall provide an Equipment Technical Manual. (CDRL A001) The Equipment Technical Manual content shall include detailed operation and maintenance information covering overall description of the equipment, descriptions of operating controls and indicators, operating instructions, general and detailed principles of operation, preventative and corrective maintenance procedures, replaceable parts lists with supporting engineering drawings/diagrams/figures.
- 3.35 Contractor shall provide a completed Statement of Volatility Worksheet for all electronic components that have data storage capability and/or removable media. The Statement of Volatility Worksheet must be provided to the Vendor.
- 3.36 Items that are not allowed onsite include but are not limited to, cell phones, Bluetooth devices, Wi-Fi devices, personal electronics, wireless transmitters, cameras, and computers. Photography is also strictly prohibited and will be enforced by the APO. If these devices are required for any particular task, exceptions must be submitted to the APO security office one week in advance of the item needing to go onsite. All submittals must be considered but are not guaranteed to be approved. The Contractor shall be a US Citizen and have no felonies, or US persons (an individual who has been lawfully admitted for permanent residence as defined in 8 U.S.C. § 1101(a)(20) or who is a protected individual as defined by Title 8 U.S.C. § 1324b (a)(3)) (2.5)

4.0 INSTALLATION:

- 4.1 All wall, floor and ceiling penetrations shall follow current ICD 705 guidance and shall be authorized by physical security Advanced Program Office (APO) and/or Information Protection Office (IPO) at time of construction. Any document referenced during the course of delivery, installation and testing shall follow the most recent and current version of said document.
- 4.2 The Contractor shall not tamper or damage fiber optic cabling during installation and the cabling shall remain fully functional at all times. If any fiber optic cabling is damaged, the Contractor shall report the damage to the SWEG Facility Engineering group immediately.
- 4.3 The piping that distributes PAO shall have a warranty against leaking for a period of five (5) years after Government inspection and acceptance of the installation. Equipment in

the mechanical rooms and labs shall be provided with a warranty for a period of one (1) year after Government inspection and acceptance.

- 4.4 Any ceiling damage caused by the Contractor at any point during the execution of this contract shall be repaired by a qualified Contractor using equivalent materials at the cost of the Contractor. The new work will be reviewed and approved by the SWEG Facilities Engineering team.

5.0 FACILITY REQUIREMENTS

5.1 Electrical

- 5.1.1 Contractor shall label all utilized breakers.
- 5.1.2 Contractor shall size wires properly according to current NEC standards.
- 5.1.3 Contractor shall provide two new Variable Frequency Drives (VFD) to control the operation of the pumps being provided by Contractor. Provided VFD's must be appropriately sized.
- 5.1.4 VFD's mounting location must be approved by facility engineer before permanently mounted.
- 5.1.5 Contractor shall utilize panel C1H2 or C1H3 located in electrical closet adjacent to lab B102 to power the pumps and chiller. Each of these panels are 3 phase 480 VAC.
- 5.1.6 Preferred motor voltage 480 VAC unless otherwise approved by facility engineer.
- 5.1.7 Control systems shall be on their own dedicated breaker.
- 5.1.8 Any cabling or wiring shall be installed in appropriately sized Electric Metallic Tube (EMT) Conduit, according to current NEC specifications.
- 5.1.9 EMT shall be mounted according to current NEC specifications.
- 5.1.10 Any copper wiring shall have a non-conductive break in the conduit when penetrating through a wall.

5.2 General:

- 5.2.1 Provide and install all material to construct overflow basin for PAO chillers and pumps.
- 5.2.2 PAO distribution piping must be mounted above drop ceiling. If distribution piping cannot be mounted above drop ceiling due to existing infrastructure this must be made known before contract is awarded.
- 5.2.3 Contractor shall submit full design for equipment and piping network within 30 days of contract award for approval by facility engineer. Drawings/details shall be submitted to the COR and SWEG PE within 30 days after award of contract. Drawings shall be in AutoCAD .dwg and/or .pdf format. Product literature shall be submitted in .pdf format. No work shall proceed without approved submittals. Government PE will have 14 days to review drawing and accept or reject. If initial submittals are rejected, Contractor shall have 14 calendar days to re-submit and Government Project Engineer shall have 7 days to review. (CDRL A001, A002)
- 5.2.4 Any changes/addendums, modification etc. to the original SOW must be approved by the Contracting Officer; no other parties have approval authority.

- 5.2.5 Contractor shall maintain a clean work area free from safety hazards throughout the entire length of the project.

5.3 Mechanical Room to House PAO Chillers

- 5.3.1 The Contractor shall construct a dedicated, sound-attenuated room at the designated location to house the PAO chillers. The room shall meet the following specifications:
- 5.3.2 Walls: Construct walls using 6-inch, 16-gauge metal studs with 5/8-inch drywall on both sides. Install R-21 insulation within the framing. Extend walls from the true floor to the underside of the metal deck above, effectively creating a full-height barrier.
- 5.3.3 Sound Attenuation: The complete room assembly (walls and doors) must achieve a Sound Transmission Class (STC) rating of 45.
- 5.3.4 Doors: Install doors with a minimum STC rating of 45. Provide at least one door of sufficient size to facilitate the passage of the PAO chillers during installation and maintenance.
- 5.3.5 Finishes: Paint the walls to match the existing laboratory aesthetic.
- 5.3.6 Floor and Ceiling Integration: The Contractor shall modify the existing raised access floor and grid ceiling as necessary to accommodate the new wall placement and chiller skid, ensuring a seamless and integrated appearance. This modification shall not be done until the mechanical room is in place and the Chiller skids arrive.

5.4 Chiller Support and Containment

- 5.4.1 The chiller room shall house the support and isolation stand(s) for the PAO chillers.
- 5.4.2 Each stand shall include a containment pan to meet the contract requirements, unless the chiller units have integrated containment features.
- 5.4.3 The new mechanical room layout must accommodate the filter unit.
- 5.4.4 The height of the support stands shall match the existing raised access floor height.
- 5.4.5 In-fill the area around the stands with raised access floor panels to maintain a consistent floor surface.

5.5 Lighting

- 5.5.1 Adjust the location of existing lighting within and around the perimeter of the new room as needed.
- 5.5.2 Supplement existing lighting with additional fixtures to ensure adequate illumination for room functionality.
- 5.5.3 Provide a dedicated light switch inside the room for independent control of the interior lighting, separate from the laboratory lighting system.

5.6 Paint/Finish

- 5.6.1 Any damage to paint must be repaired using 1 coat primer and 2 coats paint and shall match existing paint.
- 5.6.2 After removal of conduits, boxes, equipment, etc., the Contractor shall patch, paint, and repair all wall and ceiling finishes to match existing building décor. This must be evaluated for each area and the Contractor shall match finish or repaint surfaces to a corner or joint.

5.7 HVAC

- 5.7.1 Contractor shall utilize chilled water system located in the hallway adjacent to lab B102 (existing six (6) inch supply and return lines) to supply and cool the new PAO equipment.
- 5.7.2 Contractor shall be responsible for the material, equipment, and labor to provide chilled water from the chilled water system to the new PAO equipment.
- 5.7.3 Chilled water piping shall be insulated to match existing chilled water pipe insulation.
- 5.7.4 PAO piping shall be insulated.
- 5.7.5 Chilled water piping must be appropriately sized to provide enough chilled water to PAO system.

5.8 Plumbing

- 5.8.1 Contractor shall direct the overflow drain line to an overflow drain catch. This overflow drain catch must be separate from the spill catch system.
- 5.8.2 If the allocated space is not sufficiently sized for the PAO chiller system, the Contractor shall propose an alternate solution before contract is awarded.

5.9 Demolition

- 5.9.1 All material and equipment that is relocated in the course of this project shall be removed in an orderly manner. Equipment relocation shall be authorized and examined by facility engineer before and after relocation.

5.10 Preventative Maintenance

- 5.10.1 Contractor shall provide scheduled maintenance on the system as identified below:
 - 5.10.1.1 Any part numbers specified in this post work requirements can be adjusted based on system design. For example, we do not need spare parts that must not fit this system. The parts listed on this request must be filled by an item providing the same function (if applicable on this system) not necessarily the same part.
 - 5.10.1.2 Provide "Hotline" support for any unscheduled maintenance which can be performed by government personnel.
 - 5.10.1.3 Software/Hardware upgrades are at the discretion of the government and may be performed under unscheduled maintenance.
 - 5.10.1.4 Provide on-site support for more complex issues. Repair personnel need to be on-site within 3 calendar days.
 - 5.10.1.5 Any deficiencies identified during scheduled maintenance must be reported to the lab manager and addressed as the lab manager sees fit. Repairs that are deemed fixable immediately shall be accomplished at the lab manager's request.
- 5.10.2 On-going maintenance of this particular PAO Chiller System must be determined. The Contractor shall provide detailed maintenance, test and calibration results reported in hard and soft copies. Soft copies shall be delivered to the lab manager in CD format to the address 6137 Wardleigh Road, Bldg. 1515, Hill Air Force Base, 84056

5.11 Maintenance Schedule

5.11.1 Every three (3) months:

5.11.1.1 A maintenance schedule relative to on-going maintenance.

5.11.2 Every six (6) months:

5.11.2.1 Perform three (3) month maintenance schedule

5.11.2.2 Inspect and clean inlet Y strainer

5.11.2.3 Every twelve months

5.11.2.4 Perform three (3) month maintenance schedule tasks and perform six (6) month maintenance schedule tasks.

5.11.2.5 Replace Vent filter on purification unit

5.11.2.6 Check PT-21, PT-22, PT-110, PT-111, PT-210, PT-211, PT-310 and PT-311 for correct calibration. If greater than 5 PSI difference, replace unit.

5.11.2.7 Check TT-41, TT-42, TT-121, TT-122, TT231, TT-232, TT330 and TT-331 for correct calibration. If greater than 2 degrees F difference, replace unit and return failed units to OEM for Re-Calibration.

5.11.2.8 Check FT-10, AT-51 and MT-52 for correct calibration. If out of calibration, replace unit and return failed units to OEM for Re-Calibration.

5.11.2.9 Inspect hoses for cracks or signs of excessive wear and replace as needed.

5.11.2.10 Inspect all connections Inspect all connections for drips or leaks. If found replace Viton O-rings with same.

5.12 Spares

5.12.1 The Contractor shall recommend and furnish appropriate spares, including all part numbers. All such recommendations shall be subject to Government review and must be formally approved or disapproved by the Contracting Officer, or other designated authority.

5.12.2 For all part numbers, the Contractor shall recommend and supply appropriate spares. All recommendations must be approved or disapproved by the government.

6.0 PROGRAM MANAGEMENT

6.1 Management of Personnel

The Contractor shall establish processes and assign appropriate resources to effectively administer the requirement. The Contractor shall provide a list of all personnel performing under this requirement within 30 business days after receipt of award (ARO). The Contractor shall assign tasks and maintain proper and accurate time keeping records of personnel assigned to work on this requirement. The Contractor shall ensure continuation of services during extended personnel absences due to sickness, leave, and voluntary or involuntary termination from employment such that there is no negative impact to the Government mission. The CO may direct removal (permanent or temporary) of personnel for security, safety, or health reasons, upon discovery of

fraudulent credentials/qualifications, or when Contractor personnel behave in an unprofessional manner which would be considered unacceptable by a reasonable person.

6.2 Technical Requirements and Deliverables

Documentation deliverables shall be delivered IAW this SOW and the Department of Defense (DD) Forms 1423 (CDRLs) attached to the contract unless otherwise specified by the CO through a contract modification or CO letter. If there is a discrepancy in due dates between the SOW and the DD Forms 1423, this SOW shall take precedence. Due dates falling on non-working days (Saturdays, Sundays, and recognized national (legal) holidays) shall be construed to mean the next regular working day. All due dates [i.e. 30 Days ARO] are calendar counted unless designated otherwise.

The Government has the inherent right to disapprove any data item. Therefore, the Government COR must have the right to reject or require correction of any deficiencies found in deliverables. In the event of rejection of any deliverable, the COR, in coordination with the CO, must notify the Contractor in writing of the specific reasons why the deliverable was rejected. Rejected deliverables shall be resubmitted to the COR for re-inspection no later than COR suspense date. The Government must approve final versions of data deliverables.

6.3 Technical Lead

The Contractor Technical Lead shall serve as the focal point to work with the assigned Government Leads, Program Manager (PM), and/or Contracting Officer (CO) or COR. This individual typically has a detailed knowledge of daily activities associated with the SOW and is responsible for day-to-day communication with the Government Leads, PM's and/or CORs. This person shall also be a prime provider of the required services. Contractor Lead shall notify Government Leads, PMs and/or CORs daily of any manpower shortages.

6.4 Contractor Employees

Contractor shall not employ persons for work on this contract if such employee is identified to the contractor by the SWEG Security Office as a potential threat to the health, safety, security, general well-being, or operational mission of the installation and its population.

The Contractor's employees shall be required to identify themselves as contractor personnel by introducing themselves or being introduced as contractor personnel and displaying distinguishing badges or other visible identification for meetings with Government personnel. Contractor's employees shall identify themselves as contractor employees in telephone conversations and in formal and informal written correspondence.

6.5 Employee Training

The Contractor shall employ fully qualified employees with the ability to accomplish the work described in this SOW. The Contractor shall ensure all employees have been properly trained to effectively perform their assigned duties.

6.6 Federal Holidays

The following Federal holidays are observed under this contract:

New Year's Day	1 January
Martin Luther King's Birthday	Third Monday in January
President's Day	Third Monday in February
Memorial Day	Last Monday in May
Independence Day	4 July
Labor Day	First Monday in September
Columbus Day	Second Monday in October
Veteran's Day	11 November
Thanksgiving Day	Fourth Thursday in November
Christmas Day	25 December
Juneteenth	19 June

In the event that the contractor is prevented from performance as the result of an Executive Order or an Administrative leave determination that applies to the using activity, such time may not be charged to the contract. NOTE: Individual state holidays are not observed by the Federal Government and the Government facilities shall be open for work on those days. The Contractor shall plan accordingly to have employees work business as usual on those days.

6.7 Schedule

The Contractor shall respond to Government requests for contractual actions in a timely fashion. The Contractor shall adhere to all schedule timelines, milestones, delivery schedules and administrative requirements that contribute to, or effect schedule variance, to include, where needed, scheduled on-time delivery of reports, data products, billing invoices, staffing of personnel, and action items.

6.8 Business Relations

The Contractor shall successfully integrate and coordinate all activity needed to execute the task order, the timeliness, completeness and quality of problem identification, corrective action plans, proposal submittals, timely identification of issues in controversy, effective/responsive management of subcontractors, customer satisfaction, and professional and ethical behavior of the contractor management personnel.

7.0 DELIVERABLES

The Contractor shall provide and use only new commercial parts and commercial material in performance of this SOW unless otherwise authorized by the Contracting Officer. All electronics hardware must be approved before installation by the Contractor. All documents must be free from proprietary markings. All documents generated by this effort must bear the Distribution statement provided by DAF at the initial meeting. The major assemblies must be TAA compliant.

Documentation deliverables shall be delivered IAW this SOW unless otherwise specified by the CO through a contract modification or CO letter. Due dates falling on non-working days (Saturdays, Sundays, and recognized national (legal) holidays) shall be construed to mean the next regular working day. All due dates [i.e. 30 Days ARO] are calendar counted unless designated otherwise.

The Government has the inherent right to disapprove any data item. Therefore, the Government COR shall have the right to reject or require correction of any deficiencies found in deliverables. In the event of rejection of any deliverable, the COR, in coordination with the CO, shall notify the Contractor in writing of the specific reasons why the deliverable was rejected. Rejected deliverables shall be re-submitted to the CO for re-inspection no later than CO suspense date. The Government must approve final versions of data deliverables.

8.0 GENERAL INFORMATION

8.1 Place of Performance:

6137 Wardleigh Road
Bldg. 1513
Hill Air Force Base, 84056

8.2 Period of Performance

365 days after the award of the contract.

8.3 Travel

The Contractor may be required to travel on an “as needed basis” to support the objectives of this SOW. All travel (including plans, agenda, itinerary, and dates) requires prior approval by the COR. The Contractor shall provide expenditure documentation for all reimbursable cost to the COR prior to submitting an invoice in Wide Area Workflow (WAWF).

8.4 Travel Pre-Approval

The Contractor is responsible for making all necessary travel arrangements. The Contractor shall obtain written approval to travel from the Government Program Manager and COR at least five (5) business days prior to travel. Request for approval shall include the following:

8.4.1 Date and Location of Travel

8.4.2 Event (Purpose of Travel)

8.4.3 Number of Travelers

8.4.4 Name and Title (Labor Category) of Travelers

8.4.5 Goals and Desired Outcomes of Trip

8.4.6 Why Alternate Methods are not Viable Options

8.4.7 Total Estimated Cost of Trip

8.4.8 Anticipated Itinerary with Associated Estimated Costs (i.e. airfare, hotel, rental car, etc.)

8.4.9 Travel Expenses

Contractor travel policies and procedures must be best business practices IAW FAR 31.205-46, Travel Costs. The Contractor shall use FAR 31.205-46 and the DoD Joint Travel Regulation (JTR) as a guide regarding air fares, per diem, allowances and reimbursement of travel expenses. First class air travel is not authorized. Costs incurred must be IAW FAR 31.205-46. The Government reserves the right to disallow unreasonable travel costs. Travel must be billed as a separate line item on all invoices. Contractor shall submit receipts for travel expenses over \$75 to the COR IAW Federal Travel Regulation (FTR) 301-11.25.

8.5 Inherently Governmental Functions

The Contractor shall not perform any inherently Governmental functions as defined by Federal Acquisition Regulation (FAR) Subpart 7.5, Inherently Government Functions. All program decisions shall be the sole responsibility of the Government.

The Contractor shall not counsel, mentor, make judgments and/or discretionary decisions or perform any other activities related to supervision of Government personnel.

If the Contractor believes that any actions constitute or are perceived to constitute inherently Governmental functions, the Contractor shall notify the CO immediately

9.0 QUALITY ASSURANCE

9.1 Quality Control

The Contractor shall develop, implement, and maintain a comprehensive inspection system that assures compliance with all requirements of this contract IAW FAR part 46, Quality Assurance.

10.0 GOVERNMENT QUALITY

10.1 Government Provided Service Plan

- 10.1.1 When work is to be performed at the Government's location, the Government must prioritize service requirements based on operating requirements. The Government must provide the Contractor's Technical Lead a generalized projected service plan that may be reviewed or revised by the Government Program Manager as necessary. The purpose of the plan is to prioritize Government service requirements and provide the contractor with the information required to effectively manage their personnel.

10.2 Contractor Performance Assessment Reporting System (CPARS) Applicability

- 10.2.1 FAR 42.1502, Policy directs all Federal agencies to collect and report past performance information on services contracts that exceed \$1M. The CPAR assesses a contractor's performance and provides a written record on a given contract during a specific period of time. All CPARs are accomplished using the automated Navy Contractor Performance Assessment Reporting System (CPARS) found on the web at < <https://www.cpars.gov/cparsweb/home>>. This system requires the contractor to provide a designated representative to receive CPARs.

10.3 Government Inspection

- 10.3.1 Contractor performance is subject to surveillance by the COR to ensure compliance with this task order to include all elements of the Services Summary.
- 10.3.2 The contract identifies two types of non-conformances: minor and major. All records of non-conformance and corrective actions shall be documented in IAW the Quality Assurance Surveillance Plan (QASP).

10.4 Qualification Reviews

- 10.4.1 The Government reserves the option to audit the qualifications of any contractor personnel providing services to the Government under this task order.

10.5 Program Manager (PM)

- 10.5.1 The PM provides primary program oversight and management of the contract. The PM ensures the COR is trained before performing any COR duties and supports the COR's performance assessment activities. Additionally, the PM shall also review contractor's performance periodically.

10.6 Contracting Officer (CO)

- 10.6.1 The CO is the individual designated by the AF to direct and/or redirect the efforts of the Contractor or modify any of the items of this contract.

10.7 Contracting Officer's Representative (COR)

10.7.1 The COR is the Government representative appointed by the CO responsible for monitoring, evaluating and reporting Contractor performance on all applicable contracts, or under this contract. The CORs shall interface with the Customer, the Contractor, and the Program Management Office (PMO).

Appendices A-C

Appendix A ACRONYMS AND ABBREVIATIONS

AFB	Air Force Base
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
AWWA	American Water Works Association
Bldg	Building
CDRL	Contract Data Requirements List (DD Form 1423)
CS	Communications Squadron
EMT	Electric Metallic Tube (Conduit)
FY	Fiscal Year (October thru September)
GPM	Gallons Per Minute
GSSO	Government Special Access Program Officer
HAFB	Hill Air Force Base
Hazmat	Hazardous Material
IBC	International Building Code
ICD	Intelligence Community Directive
JSIG	Joint Special Access Program Implementation Guide
MCC	Motion Control Center
MOS	Months
NEC	National Electrical Code
NFPA	National Fire Protection Agency
NSCI	National Security Council Instruction
NTP	Notice To Proceed

OPSEC	Operational Security
PAO	Polyalphaolefin
PM	Program Manager
PME	
PSI	Pounds per Square Inch
PSIg	Pounds per Square Inch Gauged
RFP	Request For Proposal
SDS	Safety Data Sheet
SOW	Statement of Work
STE	
SWEG	Software Engineering Group
UFC	Unified Facility Code
USAF	United States Air Force
VAC	Volts Alternating Current
VFD	Variable Frequency Drive
SAPF	Special Access Program Facility

APPENDIX B: PUBLICATIONS / INSTRUCTIONS

The web site for Air Force Instructions (AFIs), Air Force Manuals (AFMANs), Air Force Forms (AF Form) is as follows and can be obtained from this Internet web site <http://www.e-publishing.af.mil> . The Contractor shall be responsible for researching and abiding by all applicable supplements. All other references are available via general internet searches. Every effort has been made to ensure the listed references are current as of this writing. It is the Contractors responsibility to ensure the most recent version is utilized when referenced.

Note: The above references and all applicable supplements are provided as an overview of the core AF Operation Procedures Contractor employees must be subject to and expected to follow in the accomplishment of the contract. The list is not all-inclusive.

APPENDIX C: DRAWINGS / IMAGES

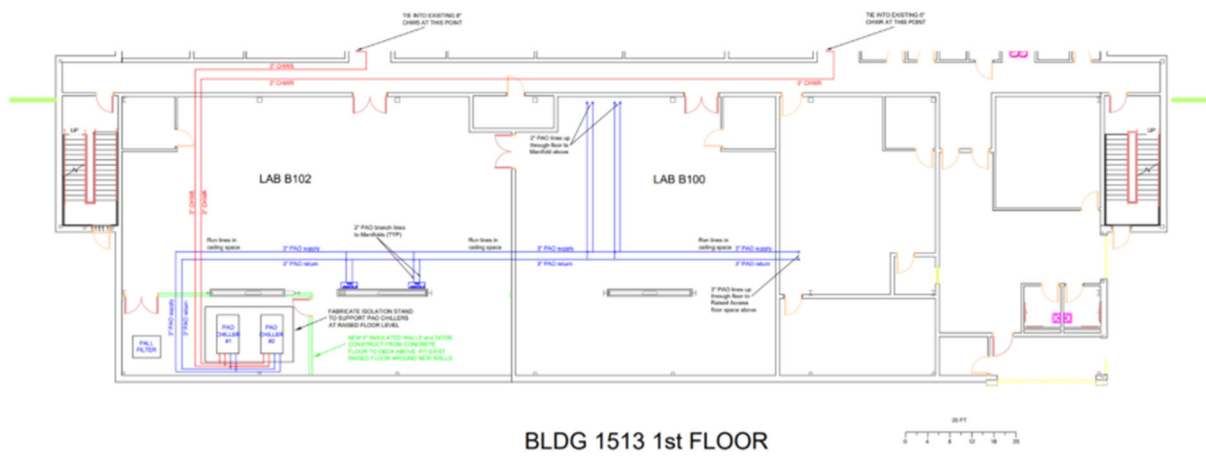
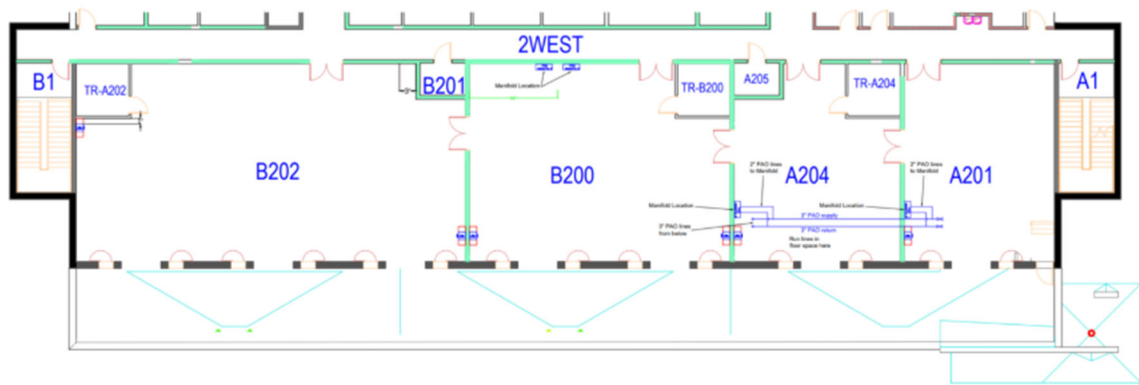


Figure 1 Building 1513 plan

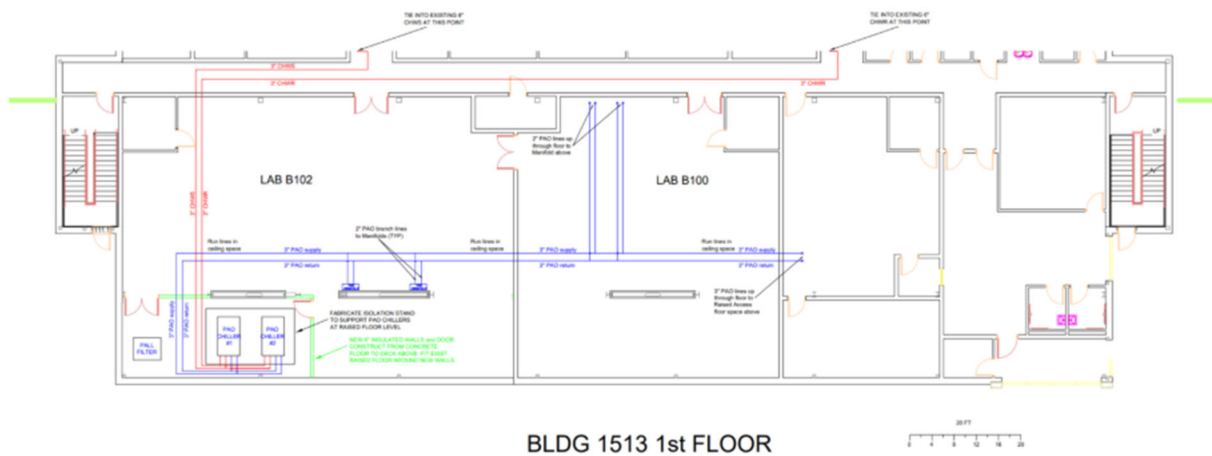


Figure 2 Building 1513 first floor

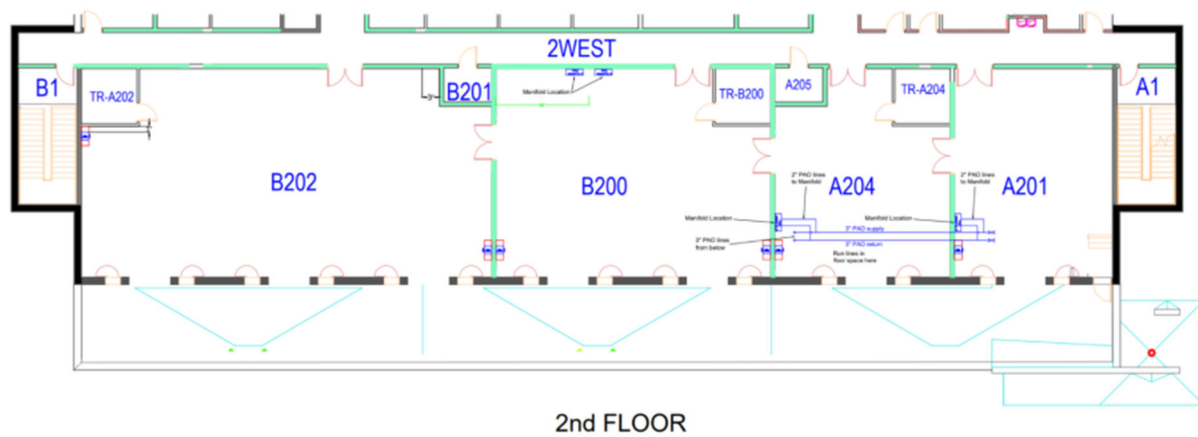


Figure 3 Building 1513 2nd floor

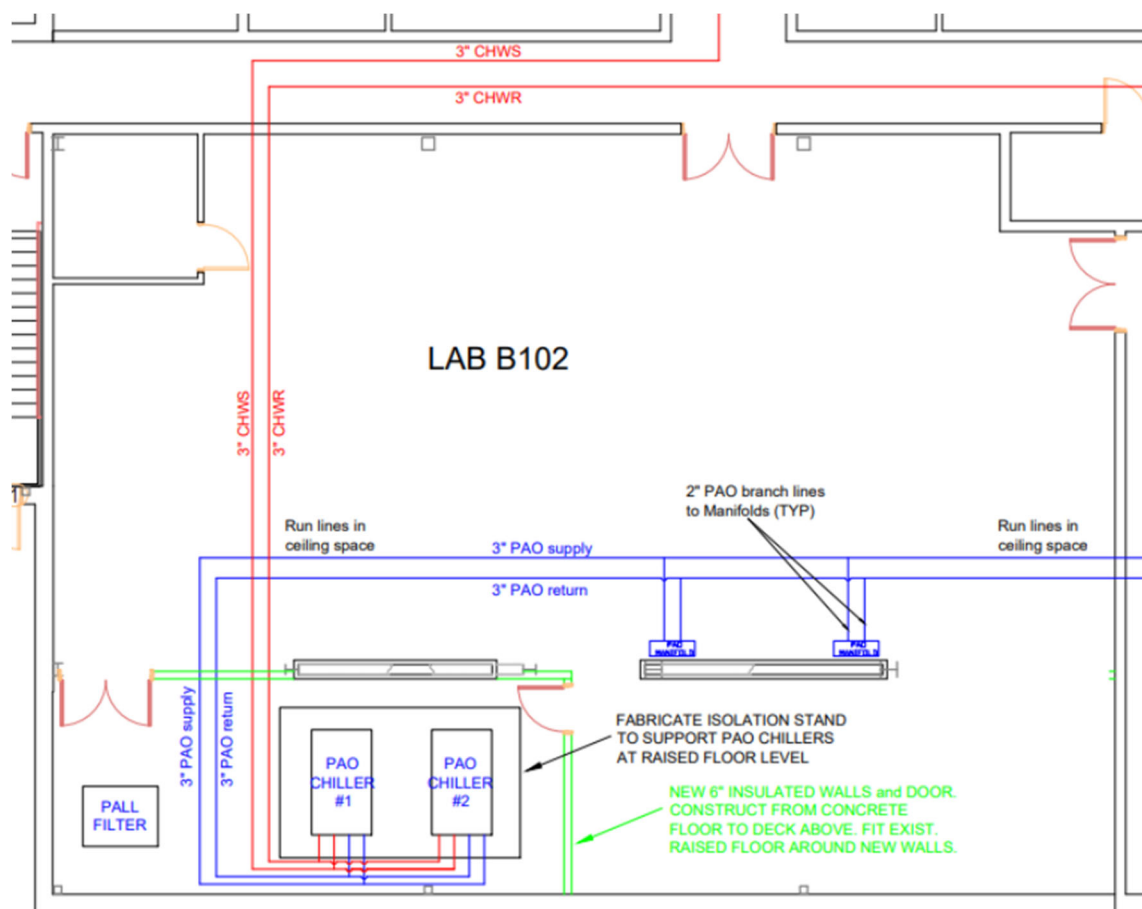


Figure 4 Room B102

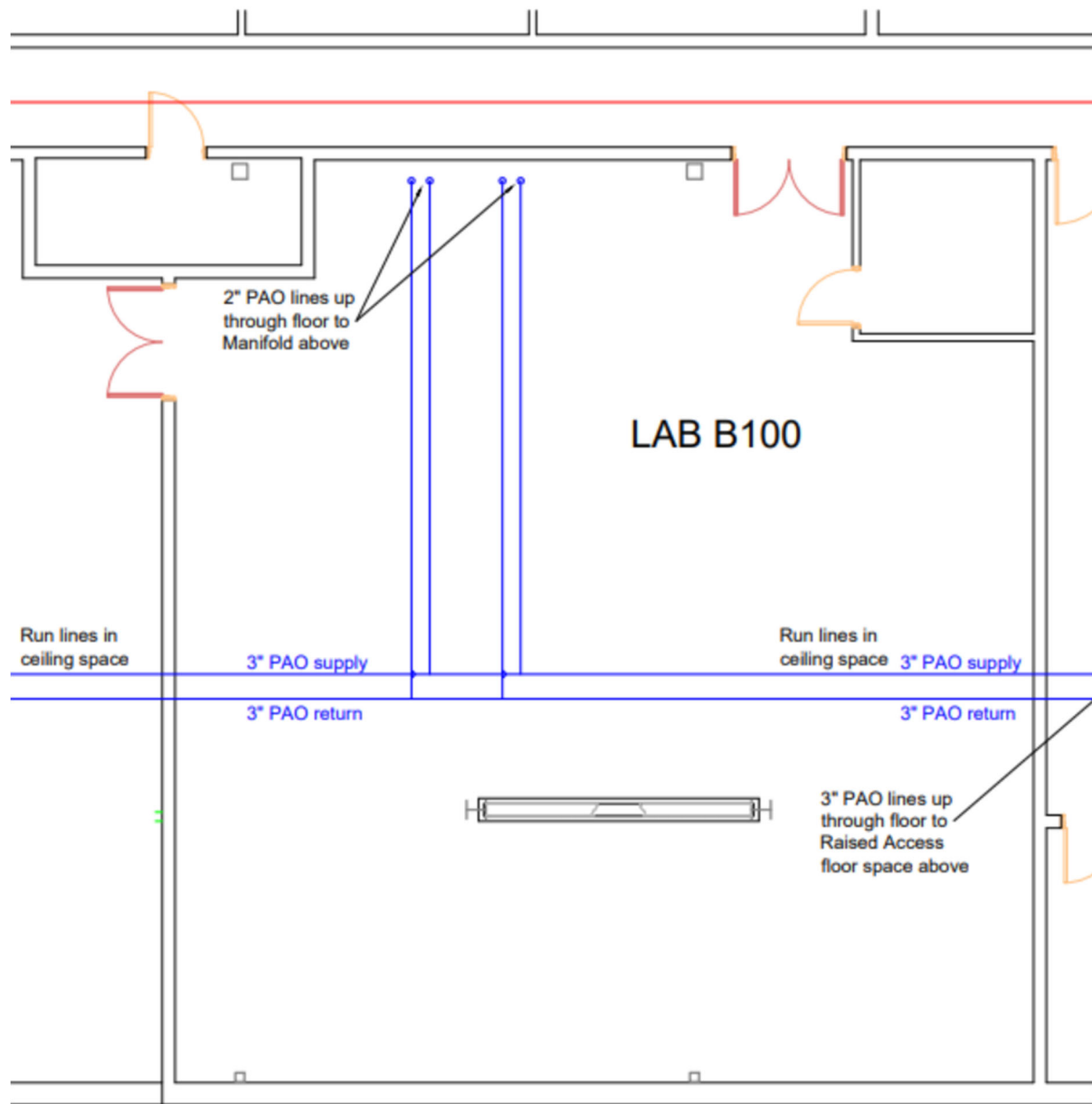


Figure 5 Room B100

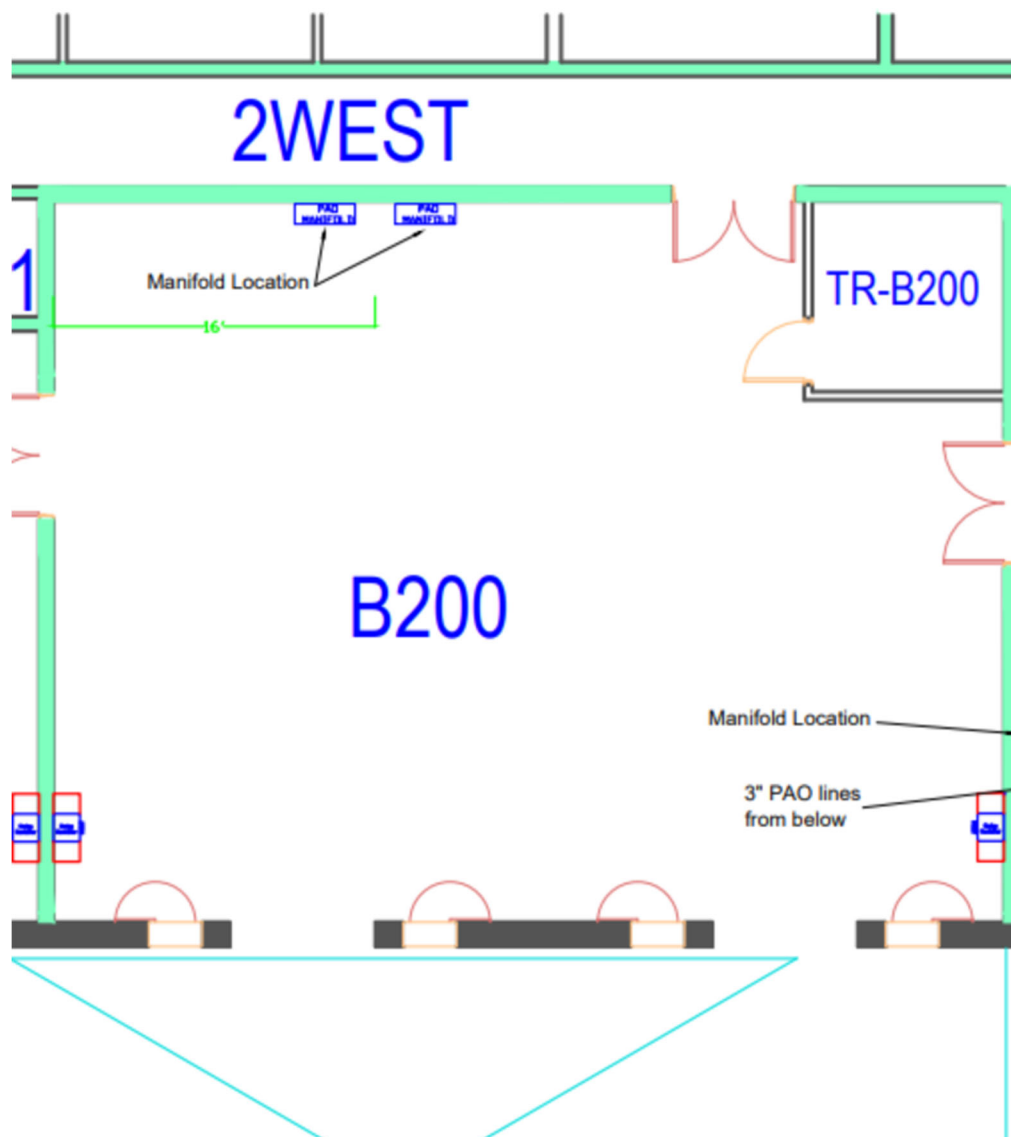


Figure 6 Room B200

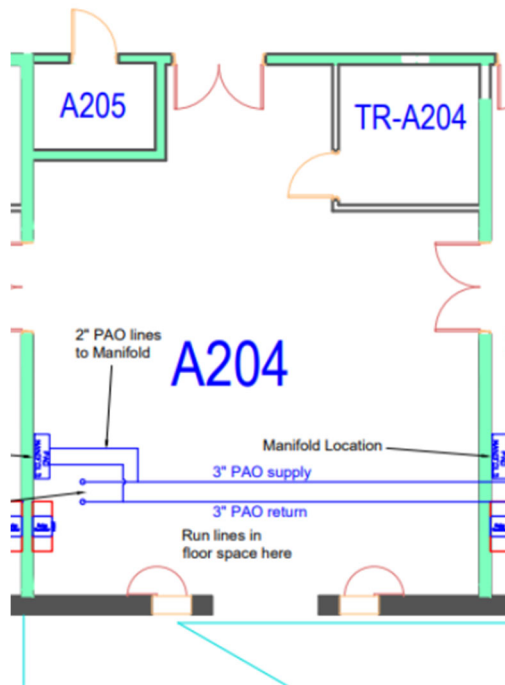


Figure 7 Room A204

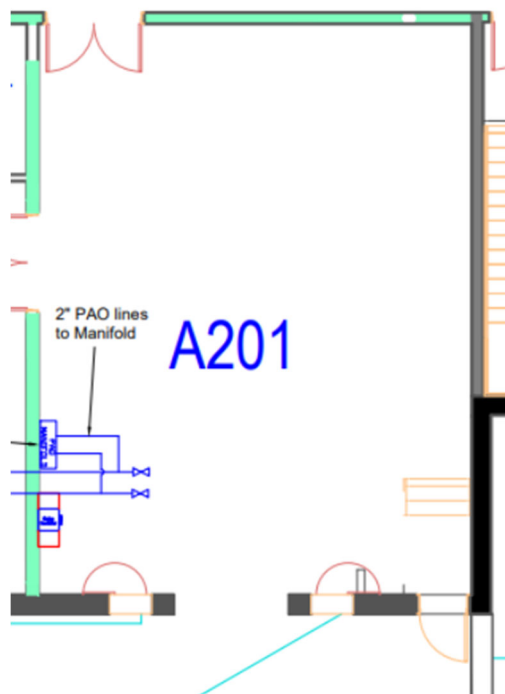


Figure 8 Room A201

Aeroquip®
3200 Series Self-Sealing Quick Disconnect



Figure 9 Aeroquip 3200 series connectors

ATTACHMENT 01
CONTRACTOR NONDISCLOSURE AGREEMENT

CONTRACTOR
NONDISCLOSURE AGREEMENT

Contract # _____

Task Order # _____

I, _____, an employee of
_____, (Contractor) in order to support the
United States Air Force (76 SWEG) and in consideration of the 76 SWEG granting
me access to Confidential, Proprietary and/or Sensitive Information held by the 76
SWEG relating to the tasks to be performed under the Task Order, intending to be
legally bound, hereby agree to the following terms and conditions:

1. I understand that my assignment to support the Contractor in its contractual
duties under the Task Order (the "Effort") shall involve access to Confidential,
Proprietary and/or Sensitive Information belonging to other contractors, other
third parties, or the 76 SWEG. The sharing of Confidential, Proprietary and/or
Sensitive Information is necessary to effectively perform the mission of the 76
SWEG.

2. Except as expressly authorized in writing by the 76 SWEG, I agree to hold
Confidential, Proprietary and/or Sensitive Information in trust and confidence and
agree that it shall be used only for my performance of duties in support of the
Contractor's performance of the Effort, shall not be used for any other purpose, or
disclosed to any third party outside the 76 SWEG. I agree that I shall not, except as
expressly authorized in writing by the 76 SWEG, at any time during or after my
performance of duties in support of the Contractor's performance of the Effort,
directly or indirectly, disclose, publish or disseminate, to any other person or entity,
any Confidential, Proprietary and/or Sensitive Information I have access to by
reason of my performance of duties in support of the Contractor's performance of
the Effort, nor shall I use any such Confidential, Proprietary and/or Sensitive
Information for my personal use or advantage, or make it available to others not
cleared for access to Confidential, Proprietary and/or Sensitive Information by the
76 SWEG or lacking a need-to-know for performance of the Effort. I shall hold all
Confidential, Proprietary and/or Sensitive Information I have access to by reason of

my performance of duties in support of the Contractor's performance of the Effort in confidence and use it solely for the benefit of the 76 SWEG and the performance of Effort. I shall not retain any Confidential, Proprietary and/or Sensitive Information, including written notes, photographs, sketches, models, memoranda or any tangible representation of the Confidential, Proprietary and/or Sensitive Information except as required in my performance of duties in support of the Contractor's performance of the Effort. Without limiting the generality of the foregoing statements, I agree that Confidential, Proprietary and/or Sensitive Information shall not be released to the media, press, competing firms, individuals, or Government officials outside the 76 SWEG for any purpose. I further understand and agree that all Confidential, Proprietary and/or Sensitive Information shall be retained, disseminated, released, and destroyed in accordance with the requirements of law and applicable Department of Defense or 76 SWEG directives, regulations, instructions, policies and guidance.

3. All information that I receive or obtain in performance of my duties in support of the Contractor's performance of the Effort shall be deemed to be Confidential, Proprietary and/or Sensitive Information, including, without limitation, visual, verbal and written information, in whatever form or medium, including, but not limited to, personal, business or financial information, or intellectual property, or technical information in the form of discussions, communications, designs, concepts, requirements, specifications, software, interfaces, components, processes, or similar things. I understand and agree that any work product related to the Effort, including, but not limited to, final documents and products, drafts, feeder documents, briefings, notes, and any other related oral or written communication shall constitute Confidential, Proprietary and/or Sensitive Information for purposes of this Agreement. Confidential, Proprietary and/or Sensitive Information shall not include information that I or the Contractor can demonstrate (A) was known by me or the Contractor without restriction as to further disclosure when received or was developed independently by the Contractor by individuals who had no direct or indirect access to such information, (B) was obtained from a source other than the 76 SWEG, its employees, agents, or contractors through no breach of confidence, (C) was in the public domain when received or thereafter enters the public domain through no breach of confidence by Contractor or myself, or (D) was used or disclosed with the prior written approval of the 76 SWEG.

4. I have been advised and understand that my violation of this agreement could cause irreparable injury to the United States. I understand that the United States Government may seek any remedy available to it to enforce this Agreement, including, but not limited to, application for a court order prohibiting disclosure of information in breach of this Agreement. I have been advised that the action can be brought against me in any of the several appropriate United States District Courts where the United States Government may elect to file the action. Court

costs and reasonable attorneys' fees incurred by the United States Government may be assessed against me if I lose such action. I understand that my violation of this Agreement may result in criminal action. I recognize that nothing in this Agreement constitutes a waiver by the United States of the right to prosecute me for any statutory violation.

5. Any questions I have regarding this agreement that I desire to direct to the 76 SWEG shall be in writing to _____

[Insert name & address of 76 SWEG POC].

6. I hereby assign to the 76 SWEG all royalties, remunerations, and emoluments that have resulted, shall result or may result from any disclosure, publication, or revelation of Confidential, Proprietary and/or Sensitive Information not consistent with the terms of this Agreement.

7. Unless and until I am released in writing by an authorized representative of the 76 SWEG, I understand that all conditions and obligations imposed upon me by this Agreement apply during the time that I am granted access to the Confidential, Proprietary and/or Sensitive Information, and at all times thereafter. Each provision of this Agreement is severable. If a court should find any provision of this Agreement to be unenforceable, all other provisions of this Agreement shall remain in full force and effect. This Agreement shall be interpreted under and in conformance with the laws of the United States.

By: _____

Print

Name: _____

Date: _____